

Prometheus: A Peek at the Popular Monitoring Tool



Prometheus is a leading monitoring solution that has seen its community grow to large numbers. The support for multiple exporters is one of the strongest points of Prometheus, since it can help you get started with specific monitoring requirements quickly.

Prometheus is a leading open source monitoring and alerting tool. It had its origins at SoundCloud and has seen significant adoption since it was announced in 2015. The software was created because of the need to monitor multiple microservices that might be running in your system. Its architecture is modular and comes with several readily available modules called exporters, which help you capture metrics from the most popular software. Prometheus is written in the Go language, and it ships with easily distributed binaries that you can use to get it running as quickly as possible.

This article looks at Prometheus, its architecture and how it can be installed. We then look at an example of monitoring your Linux node with the default support that is available out-of-the-box with Prometheus.

Prometheus' architecture

The architecture of Prometheus, taken from the official documentation, is shown in Figure 1.

The architecture might look complex but we can break it down into modules and their respective roles in the overall system. The key modules are as follows.

The Prometheus server: This is the heart of the system. This server collects the metrics from multiple nodes and stores them locally. The Prometheus server works on the principle of scraping, i.e., invoking the metrics endpoints of the various nodes

that it is configured to monitor. It collects these metrics at regular intervals and stores them locally. These metrics are pulled from nodes that run specific exporters (which are modules that extract information and translate it into the Prometheus format, which the server can then ingest). The nodes expose these over the endpoints that the Prometheus server scrapes.

Push gateway: In case the nodes are not exposing an endpoint from which the Prometheus server can collect the metrics, the Prometheus ecosystem has a push gateway. This gateway API is useful for one-off jobs that run, capture the data, transform that data into the Prometheus data format and then push that data into the Prometheus server.

Alert manager: One half of the Prometheus system is about collecting metrics. But of more importance is the ability to define your own alerts on those metrics, so that you can be notified in case of any discrepancies or levels that you might be interested in. This is the job of the alerts manager, which stores not just the alert levels but also can deliver these alerts to you over multiple channels like SMS, email, Slack, etc.

Visualisation: Prometheus comes with its own user interface that you can use to check on the configuration, nodes and graphs. Additionally, it is now compatible with Grafana, a leading open source visualisation application, so that Prometheus data is available for viewing inside Grafana. Prometheus also exposes an API, so in case you are interested in writing your own clients, you can do that too.